

Findings-first economics

Acting on these findings is modelled to improve health and reduce healthcare spending.

This report starts with what was found in the genome. The economics explains the consequence of acting on each finding; the pooled reference case shows what remains after overlapping signals, imperfect adherence and the real cost of acting are all accounted for.

INCREMENTAL QALYS

0.032

about 12 days of healthy life, on average, for a person with this pattern of findings

INCREMENTAL HEALTHCARE COST

-\$4,223

healthcare-sector perspective; negative means the modelled programme saves money

REFERENCE-CASE NMB

\$7,410

deterministic, at \$100,000 per QALY — health valued at that threshold, not cash

DECISION UNCERTAINTY

1,498 of 1,500

modelled simulations cost-effective (>99%); cost-saving in 1,434 of 1,500 simulations (96%)

8 findings carry a standardised economic estimate

4 high-confidence, 3 moderate-confidence, 2 low-confidence, 1 ungraded. 2 of the 10 findings shown are reported without a dollar figure rather than being assigned one the model cannot support.

APOE ε4 carrier

PTPN22 R620W carrier

predicted-pathogenic 17:7674220

predicted-pathogenic 12:21178615

CYP2C19 Intermediate Metabolizer

CYP2D6 Intermediate Metabolizer

SLCO1B1 Intermediate Function

Folate Metabolism (MTHFR)

1 actionable wellness variant(s)

Reproductive genetics

What rises to the top

Ranked by expected net monetary benefit, each valued on its own.
Finding-level values are not additive; the pooled reference case is calculated separately.

APOE E4 CARRIER

\$3,514

Intensive cardiovascular risk reduction (statin) to prevent MI...

MODERATE

PTPN22 R620W CARRIER

\$2,509

Awareness for autoimmune symptom clusters

MODERATE

PREDICTED-PATHOGENIC 17:7674220

\$2,341

Confirm the predicted variant clinically, then surveillance if it is...

HIGH

PREDICTED-PATHOGENIC 12:21178615

\$1,134

Confirm the predicted variant clinically, then surveillance if it is...

MODERATE

How GenomeLens avoids counting the same benefit twice

Several genetic signals can describe one underlying liability. Findings bearing on the same condition are combined on the risk scale and that condition's cost of illness is charged once, before any value is credited.

NAIVE AVOIDED COST

\$14,692

each finding's avoided cost, simply added up

condition-level pooling and adherence

→

62.8% removed — page 5 separates the overlap and adherence steps

POOLED AVOIDED COST

\$5,468

The reference-case NMB is calculated after pooling, adherence, costs and discounting. **Avoided cost and net monetary benefit are different quantities** and are not interchangeable.

Your findings at a glance

Findings that could change a decision

Every finding the model examined, the decision it bears on, the strength of the evidence, and what it is worth on its own. No dollar figure appears without the genomic result that produced it — the detail behind each figure is on pages 3 and 4.

MEDICATION & PRESCRIBING 3 findings

CYP2C19 Intermediate Metabolizer (IM) (clopidogrel)	Clopidogrel prescribing	HIGH	\$490
CYP2D6 Intermediate Metabolizer (IM) (codeine / antidepressants)	Codeine prescribing	HIGH	\$210
SLCO1B1 Intermediate Function (IM-like) (simvastatin)	Simvastatin prescribing	HIGH	\$90

RISK & PREVENTION 4 findings

APOE ε4 carrier (elevated cardiovascular / Alzheimer's risk)	Intensive cardiovascular risk reduction...	MODERATE	\$3,514
PTPN22 R620W carrier — autoimmune screening awareness	Awareness for autoimmune symptom clusters	MODERATE	\$2,509
predicted-pathogenic 17:7674220	Confirm the predicted variant clinically...	HIGH	\$2,341
predicted-pathogenic 12:21178615	Confirm the predicted variant clinically...	MODERATE	\$1,134

LOWER-CONFIDENCE & EXPLORATORY 2 findings

Folate Metabolism (MTHFR) — Nutrient-metabolism variant — deficiency...	Targeted supplementation	LOW	-\$216
1 actionable wellness variant(s)	Nutrient metabolism, sleep, and fitness...	LOW	Not yet standardised

AWARENESS — NOT MONETISED 1 finding

Reproductive genetics — 1 carrier condition(s) where partner testing...	Family Planning	NO GRADE	Not monetised
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Finding-level values are standalone expected values. They are not additive. The reference-case total is calculated separately after overlapping signals are pooled by condition.

Expected NMB values health at \$100,000 per QALY and is not cash returned to anyone. Ordering is by clinical group first and expected NMB second.

2 findings are shown without a standardised expected NMB rather than being assigned a value the model cannot support.

Medication-genotype findings

Where a genotype could change a prescribing decision

These are the clearest cases of a genomic result bearing on a future medication choice. The clinical decision leads; the economics explains why preserving the genotype for that decision has value.

CYP2C19 Intermediate Metabolizer (IM) (clopidogrel)

clopidogrel

WHAT WAS FOUND

Genotype predicts altered handling of clopidogrel.

WHY IT MATTERS

Clopidogrel is a prodrug that CYP2C19 must activate. Reduced activity means less active drug and a higher chance the antiplatelet effect is inadequate.

WHAT COULD CHANGE

Avoid clopidogrel non-response / stent thrombosis (MACE)

KEY DEPENDENCY

Value accrues only if clopidogrel is actually prescribed. The model prices that exposure explicitly rather than assuming it.

HIGH

EXPECTED NMB

\$490

Medical cost averted	\$300
Health gain	0.002 QALY
Cost to act	\$100

Expected NMB = exposure probability x adverse-event probability x relative risk reduction x cost of the avoided event, less the cost of acting. Provenance on page 8.

CYP2D6 Intermediate Metabolizer (IM) (codeine /...

codeine antidepressants

WHAT WAS FOUND

Genotype predicts altered handling of codeine antidepressants.

WHY IT MATTERS

CYP2D6 activity governs conversion of codeine to morphine and clearance of several antidepressants, so one dose can be ineffective in some people and excessive in others.

WHAT COULD CHANGE

Avoid opioid toxicity or ineffective analgesia/SSRI dosing

KEY DEPENDENCY

Value accrues only if codeine antidepressants is actually prescribed. The model prices that exposure explicitly rather than assuming it.

HIGH

EXPECTED NMB

\$210

Medical cost averted	\$120
Health gain	0.001 QALY
Cost to act	\$100

Expected NMB = exposure probability x adverse-event probability x relative risk reduction x cost of the avoided event, less the cost of acting. Provenance on page 8.

SLCO1B1 Intermediate Function (IM-like) (simvastatin)

simvastatin

WHAT WAS FOUND

Genotype predicts altered handling of simvastatin.

WHY IT MATTERS

SLCO1B1 transports statins into the liver. Reduced function raises systemic exposure and with it the risk of muscle toxicity, most clearly for simvastatin.

WHAT COULD CHANGE

Prevent statin-induced myopathy / rhabdomyolysis

KEY DEPENDENCY

Value accrues only if simvastatin is actually prescribed. The model prices that exposure explicitly rather than assuming it.

HIGH

EXPECTED NMB

\$90

Medical cost averted	\$60
Health gain	0.001 QALY
Cost to act	\$100

Expected NMB = exposure probability x adverse-event probability x relative risk reduction x cost of the avoided event, less the cost of acting. Provenance on page 8.

Decision support, not a directive. Nothing here is an instruction to start, stop or change a medication. Relevance depends on whether the drug is actually prescribed, and the prescribing decision belongs with a qualified clinician who can see the whole clinical picture.

Risk, prevention & exploratory findings

Findings with weaker evidence, reported anyway

Lower confidence changes how a finding should be read, not whether it is shown. A negative expected NMB means the modelled cost of acting exceeds the modelled benefit — useful information, not an error. Evidence grade reflects the strength of the association, not the size of the value; provenance is on page 8.

APOE ε4 carrier (elevated cardiovascular / Alzheimer's risk)

MODERATE

POTENTIAL DECISION

Intensive cardiovascular risk reduction (statin) to prevent MI — Intensive statin therapy + lipid monitoring

KEY ASSUMPTION

Baseline event probability and the effect of acting are registered assumptions varied in the sensitivity analysis; neither is measured for this individual.

EXPECTED NMB

\$3,514

medical cost averted \$3,869

health gain 0.001 QALY

cost to act \$100

PTPN22 R620W carrier — autoimmune screening awareness

MODERATE

POTENTIAL DECISION

Awareness for autoimmune symptom clusters

KEY ASSUMPTION

Baseline event probability and the effect of acting are registered assumptions varied in the sensitivity analysis; neither is measured for this individual.

EXPECTED NMB

\$2,509

medical cost averted \$2,579

health gain 0.004 QALY

cost to act \$50

predicted-pathogenic 17:7674220

HIGH

POTENTIAL DECISION

Confirm the predicted variant clinically, then surveillance if it is real

KEY ASSUMPTION

Predicted by sequence model, not observed in ClinVar — confirmation is the action, not treatment

EXPECTED NMB

\$2,341

medical cost averted \$1,713

health gain 0.014 QALY

cost to act \$0

predicted-pathogenic 12:21178615

MODERATE

POTENTIAL DECISION

Confirm the predicted variant clinically, then surveillance if it is real

KEY ASSUMPTION

Predicted by sequence model, not observed in ClinVar — confirmation is the action, not treatment

EXPECTED NMB

\$1,134

medical cost averted \$1,055

health gain 0.009 QALY

cost to act \$0

Folate Metabolism (MTHFR) — Nutrient-metabolism variant...

LOW

POTENTIAL DECISION

Targeted supplementation (vitamin D/B12/folate/iron)

KEY ASSUMPTION

Baseline event probability and the effect of acting are registered assumptions varied in the sensitivity analysis; neither is measured for this individual.

EXPECTED NMB

-\$216

medical cost averted \$219

health gain 0.001 QALY

cost to act \$100

Reported without a standardised value: 1 actionable wellness variant(s), Reproductive genetics — 1 carrier.... No registry-backed economic pathway exists for these yet, so none is invented. They appear with their decision and evidence grade on page 2.

How the findings combine

Taken together, what do these findings mean?

This is the canonical reference case. Findings bearing on the same condition are combined on the risk scale, each condition's cost of illness is charged exactly once, and trial efficacy is discounted to real-world adherence before any value is credited.

INCREMENTAL
HEALTHCARE COST

-\$4,223

healthcare-sector perspective

INCREMENTAL QALYS

0.032

about 12 days of healthy life on average

REFERENCE-CASE NMB

\$7,410

at \$100,000 per QALY

CLASSIFICATION

dominant (more health, lower cost)

not defined — strategy dominates or is dominated

Condition pools — combined before value is credited

CONDITION	FINDINGS	BASELINE	RAW	POOLED	ADHERENCE	REALISED	COST AVERTED	QALYS	NMB
Pathogenic	3	20.0%	94%	52%	65%	36%	\$3,705	0.031	\$6,273
Alzheimer	1	15.0%	30%	30%	35%	11%	\$1,630	0.001	\$1,516
CAD	1	20.0%	3%	3%	50%	2%	\$132	0.0004	-\$79

Two corrections, applied in order

The engine reports these as one combined reduction. They rest on different evidence and are separated here: overlap is a structural correction, adherence a behavioural one.



Adherence is charged to the benefit *and* to ongoing intervention cost, but not to the one-off test cost. 0.016 QALYs are lost to imperfect adherence relative to trial efficacy.

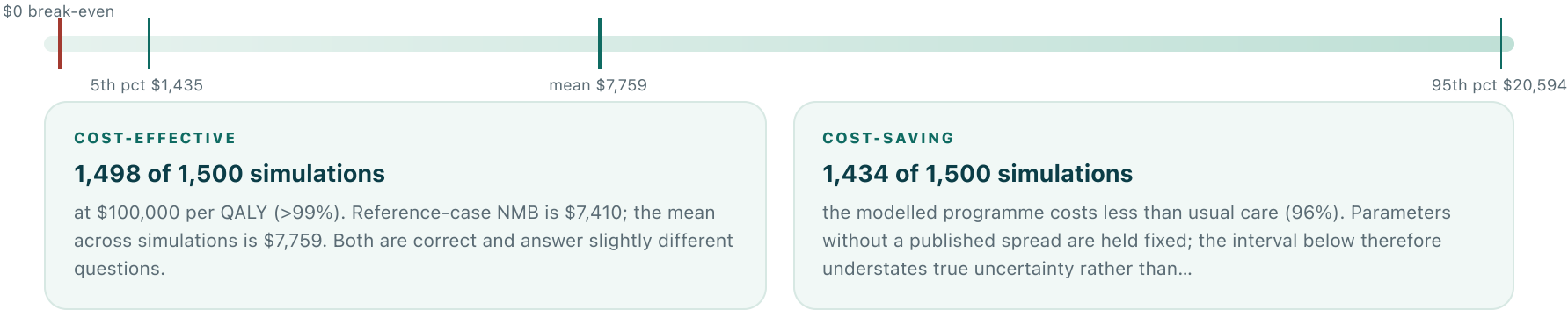
\$7,410

Uncertainty, evidence & model discipline

How sure is this, and what rests on judgement?

Uncertainty has two layers: the spread of modelled outcomes, and the provenance of the inputs producing that spread. A robust decision does not make its inputs precise, so both are shown.

Net monetary benefit across 1,500 simulations



What drives the answer

baseline event probability ASSUMPTION		\$12,229
coi pathogenic generic DERIVED		\$10,004
actionable rrr ASSUMPTION		\$5,748
marginal cost fraction DERIVED		\$3,645
baseline event probability dementia ASSUMPTION		\$3,608
adherence screening ASSUMPTION		\$3,424

Swing in net monetary benefit across each parameter's plausible range. Amber bars are declared assumptions rather than sourced values; they account for 62.2% of total swing.

EVIDENCE FOUNDATION

Registry sourced	49 / 65 · 75.4%
Whole model — PMID/DOI	46.9%
Whole model — attributed	95.8%
Whole model — judgement only	4.2%

Two different denominators — 65 registered parameters and 382 known model parameters. Reported separately, because collapsing them into one figure would create an inconsistency rather than remove one.

WHAT COULD CHANGE THE ANSWER

Most of what drives this answer is guesswork, not measurement. The biggest single driver is how much acting on a genetic risk actually reduces it — a number nobody has measured directly for most of these findings. Treat the figures as a considered estimate, not a forecast.

0 ERRORS report-consistency validation — rendering permitted

5 warnings · 11 information notes, listed in full in the technical appendix

The testing decision

Is whole-genome sequencing worth it?

Two different questions live on this page and they are never added together. One is a choice made *before* testing, averaged over a population. The other counts what sequencing actually found in this genome.

PROSPECTIVE — BEFORE TESTING		OBSERVED — IN THIS GENOME	
What is the expected value of choosing sequencing rather than an array?		What did sequencing actually find that an array could not?	
Chance an array misses an actionable finding	1 in 56	Sequencing-only findings present	2
Value if you are that person	\$25,189	Observed standalone contribution	\$3,475
Expected value, averaged over everyone	\$491	A count of findings this genome contains, not an expectation. For this input the population question is already settled.	
Extra cost over an array	\$200		
Net expected value	\$291		

Separation rule. The prospective figures describe the population this genome was drawn from; the observed figure describes the genome itself. They answer different decision questions, come from different inputs, and are never combined into one headline total.

Strategy frontier

STRATEGY	COST	QALYS	ICER VS PREVIOUS	STATUS
No testing	\$0	20.000	—	efficient
Genotyping chip	\$200	20.060	\$3,333/QALY	efficient
Whole-genome sequencing	\$600	20.120	\$6,667/QALY	efficient
Targeted clinical panel	\$900	20.090	—	dominated

Frontier values illustrate the testing choice for a population and are not this genome's reference case. This is an average over a population, and the distribution is extremely skewed: the great majority of people get nothing, and a small number get something that changes their care substantially. An expected value is the right...

What was assumed, where it came from, what was checked

The first seven pages tell the story; this page is their audit trail. Full parameter tables, references and the advanced analyses follow in the technical appendix.

MODEL		PROVENANCE	
Perspective	healthcare sector	Registered parameters	65
Comparator	usual care	Published	8
Outcomes	cost, QALYs, NMB	Derived	41
Time horizon	10 years	Declared assumptions	16
Discounting	3%, costs and QALYs	Registry sourced	75.4%
Willingness to pay	\$100,000/QALY	Model, resolvable	46.9%
Currency	USD, United States	Model, attributed	95.8%
Pooling	complement of products	Model, judgement only	4.2%
Adherence	benefit and ongoing cost	A parameter declared as an assumption may not cite a source — the registry refuses to load if one does.	
Uncertainty	1,500-run PSA		

Declared limitations

- Population-average parameters are applied to one genome. The model values expected consequences for a person with this pattern of findings, not a prediction for any individual.
- Estimates are illustrative and decision-analytic. This is not a formal economic evaluation or a submission to any authority.
- Where a condition-specific baseline risk is unavailable a generic registered assumption is used; those assumptions dominate the sensitivity analysis and are labelled as such.
- Costs are not inflation-normalised to a single price year.
- Findings without a registry-backed pathway are excluded from net monetary benefit rather than assigned an invented value.
- Clinically relevant findings require confirmation by an appropriate professional before anything follows from them.

Validation

- PASS** NMB identity — $\lambda \times \Delta QALY - \Delta Cost$ reconciles to the reported total

PASS Condition cost charged once — no pool exceeds its cost of illness

PASS WGS basis separation — observed and prospective never combined

PASS Report validation gate — no errors; rendering permitted
- PASS** Dominance classification — sign of cost and effect agrees with the label

PASS Pooled risk cap — combined risk reduction stays within its bound

PASS Budget-impact consistency — peak burden is the maximum net spend

Audit trail. Every figure in this report is serialised in `economics-payload.json`, the canonical object these pages render from. The parameter registry, references, cross-path reconciliation and the advanced analyses follow in the technical appendix. Build GENOMELENS-BUILD: 8205ac1+dirty 2026-08-29T13:03:34-04:00. Synthetic input — no human genome was used.